

Information for First and Second Responders

Emergency Response Guide



Kalmar Ottawa OT2 EV Electric Terminal Tractor

2025 to Present



Photos



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Version
No.

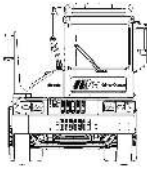
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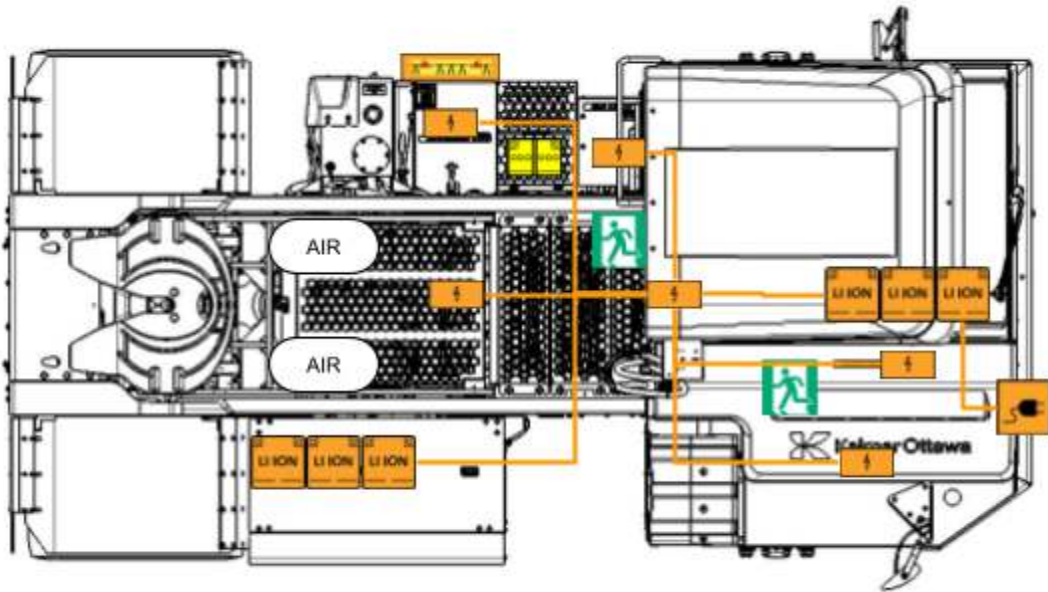
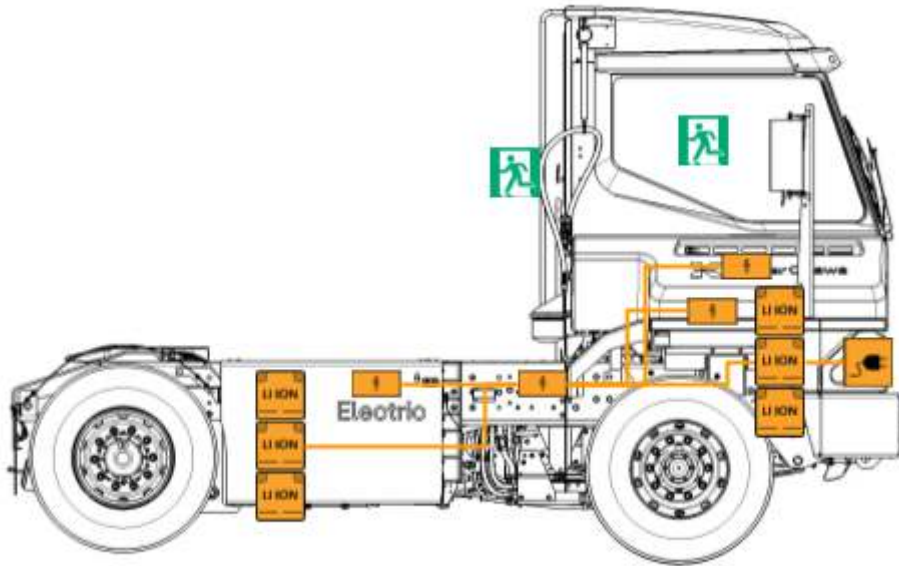
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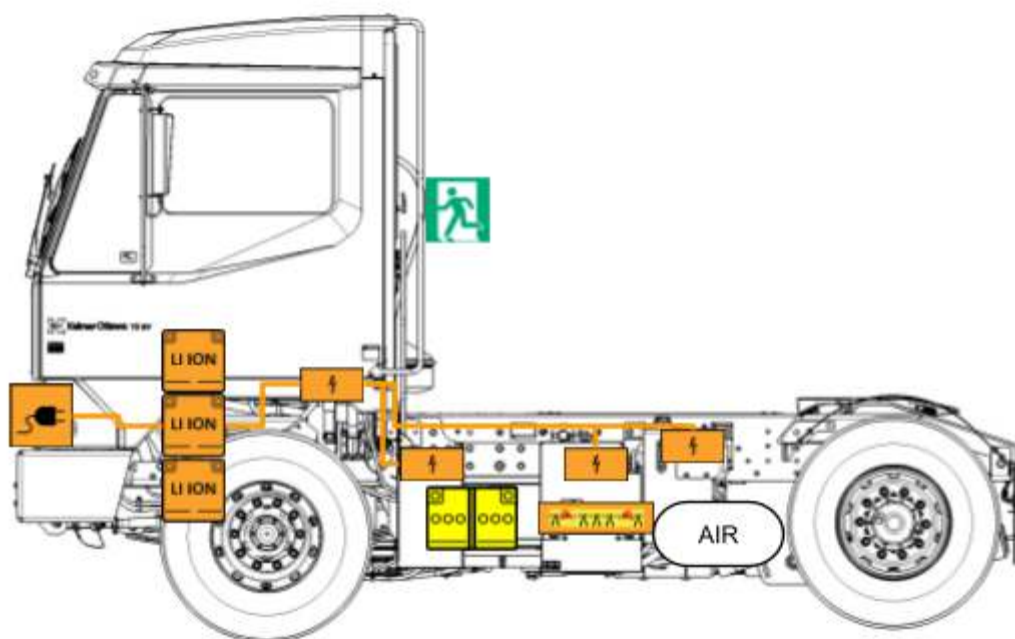
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Front page of rescue sheet

	Kalmar Ottawa OT2 EV Electric Terminal Tractor 2025 to Present		LHD pictogram 
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Rescue sheet illustration





Legend

Electric Propulsion	Charge Port	High-Voltage Li-Ion Battery	High Voltage
Low-Voltage Battery	Exits	High-Voltage Power	Cable Cut High Voltage

WARNING

If high-voltage equipment or high-voltage cables (orange sheathing) are damaged due to an accident related to the equipment shown above, there may be a short circuit. Be sure to put on insulated protective gear, such as insulated clothes and gloves, before starting rescue operations.

NEVER CUT HIGH-VOLTAGE CABLES (ORANGE SHEATHING)

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NOTE Items for consideration under the respective heading are given in the following pages.

1. Identification / recognition

WARNING

Lack of noise does not mean the vehicle is off. Silent movement or instant restart capability exists until the vehicle is fully shut down.
Wear appropriate personal protective equipment (PPE).

Note: The “**EV**” in the model designation indicates a fully electric vehicle
The model logo is located on the driver’s side of the cabin. The model logo is also located on the front service panel.

2. Immobilisation / stabilization / lifting

Raising a Vehicle

1. Set the parking brakes, and shut down the vehicle. Chock the tires.

WARNING

Do not use bottle jacks to raise the vehicle.
Bottle jacks can slip, allowing the vehicle to fall, which could result in damage to the vehicle, serious injury, or death.

NOTICE

Do not place jack stands under any of the suspension components; doing so could cause suspension component damage. Jack stands can be placed at any point below the axle, including the differential area. Do not lift the vehicle from the batteries. See Fig. 1 below for an illustration of no lift areas.

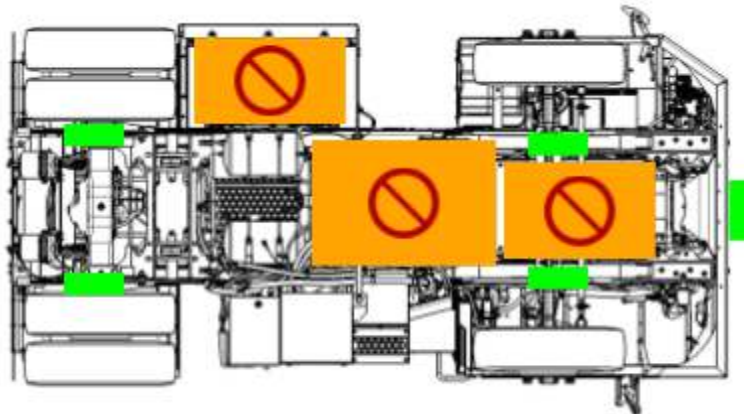


Figure 1

IMPORTANT: Only lift unloaded vehicles disconnected from trailers.

2. Place a floor jack under the axle mounting plate.

3. Raise the vehicle. Add additional jack stands under the axles as needed to support the vehicle.

3. Disable direct hazards / safety regulations

In case of fire, submersion, accident, or other emergency, disable the high-voltage system. To disable the high-voltage system:



Emergency Responder Cable Cut Point

- Double cut the HVIL cut loop to break the high voltage interlock loop telling the vehicle that there is a high voltage connector open. The vehicle will respond by opening the contactors, disabling high voltage to accessory components.
- A cable cut point is located in the low voltage box next to the driver's side frame access steps. A cable cut point is shown in Fig. 2 below.

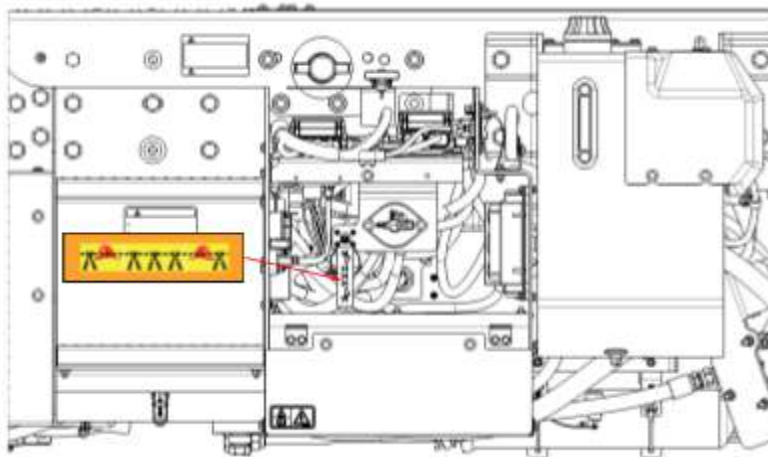


Figure 2

WARNING

Not every high voltage component is labeled. Always wear appropriate PPE. Always double cut the first responder loop. Do not attempt to open the High Voltage (HV) battery.

4. Access to the occupants

- The main access door is located at the rear of the cab.

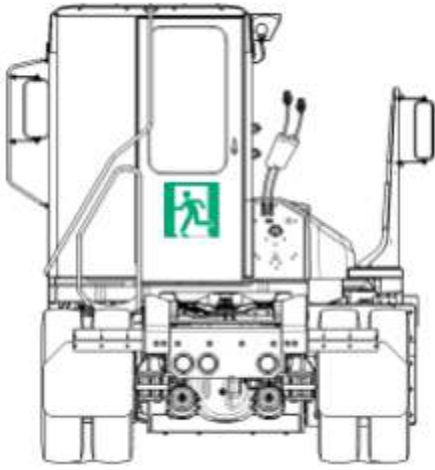


Figure 3

- Some vehicles are equipped with an optional access door on the driver's side of the cab.

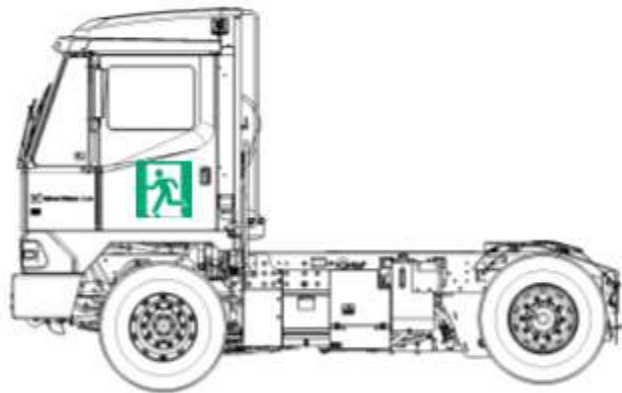


Figure 4

- The right side window is equipped for emergency glass break as an emergency exit. There is a glass break tool located on the transmission shifter panel next to the window to be used to break the glass.

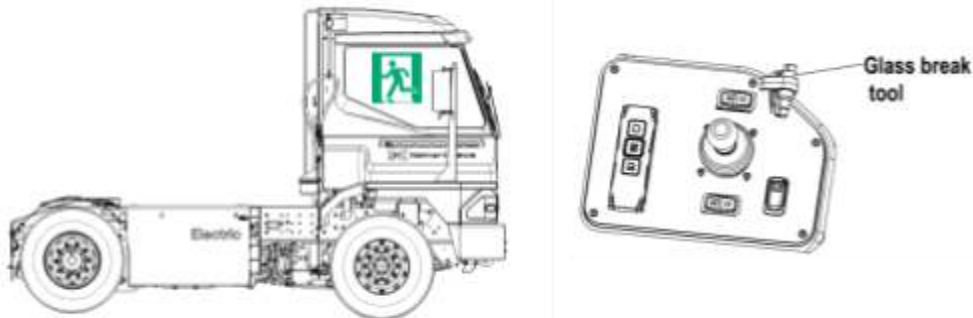
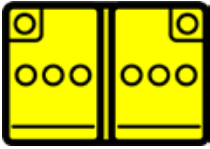


Figure 5

5. Stored energy / liquids / gasses / solids

	  	12V
	     	800V

WARNING

The cells in the HV battery are sealed and there is not enough electrolyte to create a pool of liquid. Clear liquid is likely water. The coolant is red.

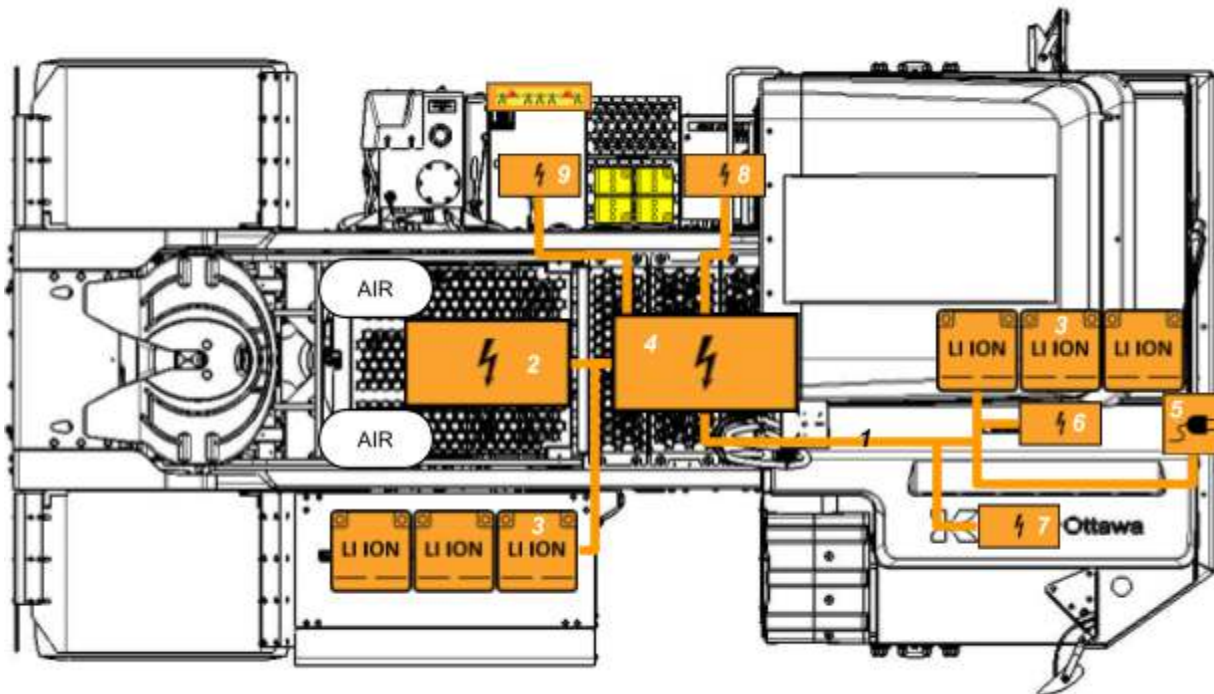


Figure 6



High Voltage Components

1. High voltage cabling
2. Traction motor
3. High voltage battery packs
4. High voltage stack
5. Charge port
6. Thermal management system
7. Air compressor
8. Hydraulic motor
9. DC-DC converters



High Voltage Battery Pack

The OT2 EV is equipped with three (3), four (4), or six (6) lithium-ion high voltage battery packs providing a system nominal voltage of ~610V. Each pack is either 200V (3 or 6 pack system) or 155V (4 pack system). The battery is made up of many cells that are liquid cooled with coolant. The coolant will appear red in color and may leak from the battery pack if the pack has been compromised during a vehicle collision. The battery cells will have stored energy within them. Never breach the high voltage battery when lifting from under the vehicle. Refer to Chapter 2: Lift Areas for instructions on how to properly lift the vehicle.



High Voltage Power Cable / Component

High voltage cables are shown in orange. There are high voltage cables that run the length of the battery pack. At no time should any high voltage cables be compromised with rescue tools. The assumption should be made that at all times there is high voltage present in the Orange High Voltage Cables.

AIR

Air tanks

The assumption should be made that contents are under pressure. Proceed with caution if cutting air lines. Relieve pressure in air tanks when possible. See Fig. 7 below for reference and use air tank drain cables to relieve pressure.

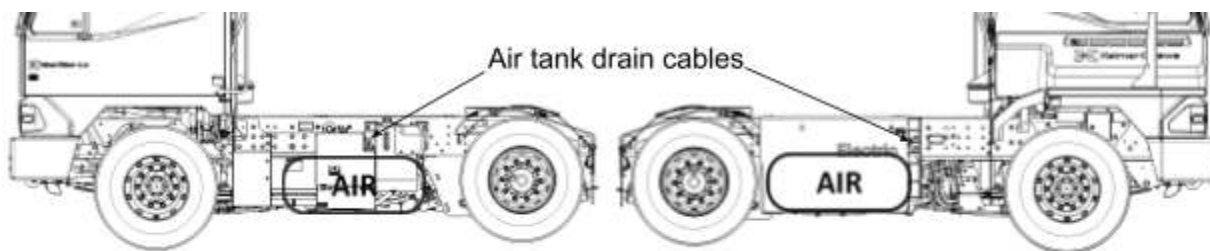


Figure 7

6. In case of fire

Firefighting

		
	DO NOT SUBMERGE VEHICLE TO EXTINGUISH / COOL BATTERY FIRE	 USE LARGE AMOUNTS OF WATER
	POSSIBLE BATTERY RE-IGNITION!	
		MONITOR HV BATTERY TEMPERATURE FOR AT LEAST 24 HOURS

Lithium-ion batteries contain liquid, flammable electrolyte.

Burning batteries can also ignite other batteries in the vicinity.

The extinguishing agent must be applied continuously until fully cooled down; otherwise, there will be a risk of a new ignition.

- A burning lithium-ion battery generally cannot be extinguished directly. Water as the extinguishing agent can be used for cooling lithium-ion batteries.
- A battery fire may continue to burn for several hours or re-ignite, so it is recommended to continue to cool the battery with excessive amounts of water.
- The temperature of the battery can be monitored with a thermal imaging camera to ensure it is not heating up.
- Re-check the temperature of the battery once an hour to ensure the temperature is not increasing. If the temperature is increasing, continue to cool the battery with water.

Do not store a vehicle containing a damaged or burning lithium-ion battery within 50 feet (15m) of a structure or other vehicles.

WARNING

Always wear appropriate PPE when fighting vehicle fires.

Responders should always protect themselves with full PPE, including SCBA. If the High Voltage Battery is burning, PPE that protects from electrical shock shall be used.



USE WATER TO FIGHT A HIGH VOLTAGE BATTERY FIRE. If the battery catches fire, is exposed to high heat, or is generating heat or gasses, apply large amounts of water directly to the battery to cool the battery. Always establish or request additional water supply early. Do not open the battery pack for the purpose of cooling it.

Placing the vehicle in a large container full of water is not recommended. The use of a Thermal Imagery Camera or Infrared (TIC or IR) is recommended to monitor battery temperatures during the cooling process. Continue to use water until the battery has reached ambient temperatures or below, indicated by the thermal imagery camera. When utilizing a thermal imaging camera, allow enough time, once the application of water has stopped, to allow for heat within the battery to transfer to the battery enclosure.

Extinguish small fires that do not involve the high voltage battery using typical vehicle fire fighting procedures.

During fire extinguishing, do not make contact with any high voltage components. Always use insulated tools for fire extinguishing.



Battery fires can take up to 24 hours to fully cool. After suppression and smoke has visibly subsided, a thermal imaging camera can be used to actively measure the temperature of the high voltage battery and monitor the trend of heating or cooling. There must be no fire, smoke, audible popping/hissing, or heating present in the high voltage battery for at least 45 minutes before the vehicle can be released to second responders (such as law enforcement, vehicle transporters, etc.). The battery must be completely cooled before releasing the vehicle to second responders or otherwise leaving the incident.

Always advise second responders that there is a risk of battery re-ignition. Second responders should be advised to position the vehicle to drain excess water out of the vehicle by tilting or repositioning it. This operation can assist in mitigating possible re-ignition.

Due to potential re-ignition, an OT2 EV that has been involved in a submersion, fire, or a collision that has compromised the high voltage battery should be stored in an open area at least 50 feet (15 m) from any exposure.

High-Voltage Battery — Fire Damage



Similar to conventional and other electric and hybrid vehicles, a burning battery releases superheated gasses and toxic vapors. This release may include volatile organic compounds, hydrogen gas, carbon dioxide, carbon monoxide, soot, particulates containing oxides of nickel, aluminum, lithium, copper, cobalt, and hydrogen fluoride. Responders should always protect themselves with full PPE, including SCBA, and take appropriate measures to protect civilians downwind from the incident.

The high voltage battery consists of lithium-ion cells. If damaged, only a small amount of fluid can leak.

The high voltage battery and drive unit(s) are liquid cooled with a typical glycol-based automotive coolant. If damaged, this red coolant can leak out of the high voltage battery.



A damaged high voltage battery can create rapid heating of the battery cells. If you notice smoke, steam, or audible popping or hissing coming from the high voltage battery, assume that it is heated and take appropriate action as described above.

7. In case of submersion

DANGER

Handling a submerged vehicle without appropriate training and personal protective equipment (PPE) can result in serious injury or death.
The removal and de-energizing of a partially or completely submerged vehicle should always be handled by trained emergency responders outfitted with the required PPE.

WARNING

Do not disable the high-voltage system by cutting through an emergency responder cable cut point if the cable is submerged.
Submersion in water (especially salt water) can damage low and high-voltage components. Cutting a submerged cable can result in an electrical short and subsequent fire once the vehicle is no longer submerged. This could cause serious injury or death.

WARNING

Damaged high-voltage batteries can produce flammable gas and fire which can lead to serious injury or death.
Vent the passenger compartment once the vehicle is out of the water.
Do not store the recovered vehicle indoors.

The high-voltage system of the OT2 EV is isolated from the chassis. When undamaged, the system will not energize the surrounding water, even when fully submerged.
If you are in a vehicle that becomes partially submerged, exit the vehicle quickly and safely. To minimize risk, avoid contact with a submerged high voltage system and batteries. If you have time, turn the key off on the dash prior to exiting.
When you are in a safe location, immediately contact emergency services.
Emergency responders will check for damage and, after removing the vehicle from the water, disable the high-voltage system.
Vehicles that have been submerged in water should be handled with greater caution due to the potential risk of a high voltage electrical battery fire. First Responders should be prepared to respond to a potential fire risk. After the vehicle is removed from the water, continue normal disabling procedures as outlined in Chapter 3.

8. Towing / transportation / storage

Towing

WARNING

Do not tow an unbraked vehicle if the combined weight of both vehicles is more than the sum of the gross axle weight ratings (GAWR) of the towing vehicle. Otherwise brake capacity will be inadequate, which could result in personal injury or death.

Towing a vehicle requires special equipment and training. Kalmar recommends that a professional towing service be used when towing a disabled Kalmar tractor. The best way to tow a Kalmar tractor is with the rear wheels lifted off the road. This prevents any possible damage to the drivetrain and high voltage systems. Towing the vehicle with the rear wheels lifted avoids having to disconnect the driveline. Also, towing with the rear wheels lifted does not require the spring brakes to be caged. If it is impossible to tow the vehicle with the rear wheels lifted be sure to follow the steps listed below, "Towing Vehicle with Front Wheels Suspended".

NOTICE

Some vehicles are equipped with "Off Highway" tires. Tire damage may occur if towed on the road at speeds greater than 25 mph

Towing Vehicle with Rear Wheels Suspended

The recommended way to tow a disabled vehicle is with the rear wheels lifted off the ground. The steering axle must be locked in the straight-ahead position. (See Manually Releasing Tractor Spring Brakes (Caging) under 5.4.4 Parking brake system)

WARNING

Failure to protect high-voltage (HV) cables from towing chains could cause property damage, severe personal injury, or death.

Towing Vehicle with Front Wheels Suspended

It is not recommended to tow a vehicle with the front wheels lifted and the rear drive wheels on the road. This practice may result in serious vehicle damage. Pay close attention to the following rules to prevent vehicle damage if you must tow your Kalmar tractor with the rear wheels on the road.

WARNING

Always obey the following when towing a Kalmar tractor with the front wheels lifted and the rear wheels on the road. Failure to do so could result in death or serious injury.

1. Always use a rigid towing bar or properly restrain the towed vehicle. Using a chain or cable to tow the vehicle is not recommended.
2. Always disconnect the driveline to the rear drive axle(s).
3. Always cage the spring brakes on all rear drive wheels. A loss of air pressure could

occur while towing the vehicle. This would apply the spring / parking brakes and lock the rear wheels on the towed vehicle

⚠ WARNING

Do not exceed the maximum speed rating of the OT2 EV. Failure to limit speed will result in damage to the rear axle.

⚠ WARNING

Failure to disconnect the driveline before towing will result in generation of high voltage that will cause equipment damage and may lead to injury or death.

⚠ WARNING

When manually releasing the spring brakes make sure that the vehicle wheels are properly blocked. If the wheels are not blocked the vehicle could move suddenly when the spring brakes are released and cause death or severe injury.

⚠ WARNING

If the disabled vehicle is connected to a tow vehicle before the spring brakes are released make sure that the tow vehicle's parking brakes are applied and its wheels are blocked to prevent movement. Failure to do so could result in death or serious injury.

Use of Tow Hooks

Tow hooks are not designed for on-road towing; they should be used to recover and move the vehicle to a position where it can be hooked up properly for towing.

NOTICE

When using tow eyes at the rear or front of the vehicle to move the vehicle, do not pass a sling (for example, a rope or chain) from one loop to another. Known as reeving, this practice is not permissible in most industrial applications of towing and hoisting. Reeving can overload the loops and result in damage to the vehicle. See example Fig. 8 below.



Figure 8

Lifting

Attach the lifting slings to the front bumper and to the lifting ears at the rear section of the frame. When lifting, select the length of the lifting slings so that the tractor is balanced both lengthwise and crosswise. Lifting slings can not touch the windscreen when lifting. Pay special attention to protecting the cab and attached components to prevent damage to them while lifting.

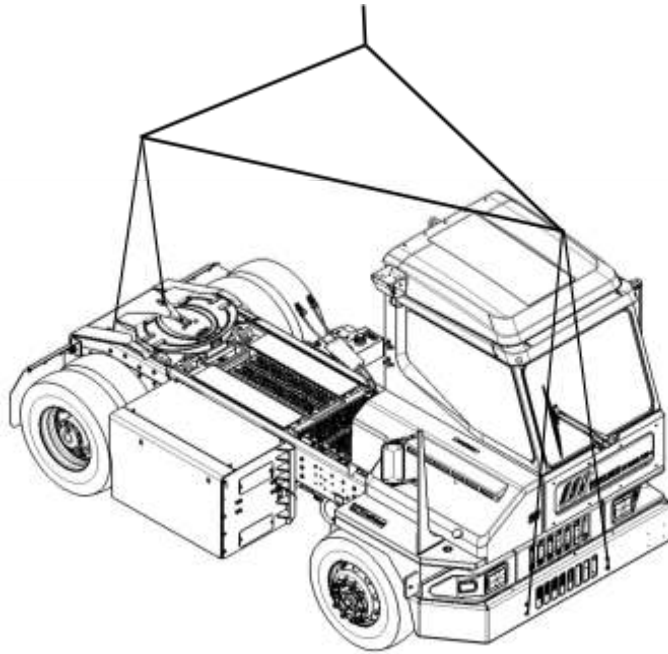


Figure 9

DANGER

Never go under a suspended load while the machine is lifted.

Transport

Engage the parking brake and secure the tractor to the cargo bed with, for example, chains so that the load cannot shift and cause hazard to other road users and the tractor remains intact during transportation.

WARNING

Failure to check load height may result in the truck striking bridges or other overhead obstacles. It may be necessary to remove roof mounted accessories.

Storage

WARNING

Possible batter re-ignition! After a fire incident, store outside at a safe distance (50 FT/ 15 M) from other vehicles and structures!

Storage of Damaged Battery

1. If the lithium-ion battery has been damaged, it is possible that the battery can increase in temperature and lead to a fire. Use a thermal imaging camera to ensure that battery is not increasing in temperature or above 60° C. Take a measurement once an hour to verify if the temperature has stabilized or is increasing. If needed, initiate cooling with water.
2. Before handling the damaged battery ensure there is no smoke or signs of heat. If after observing the battery pack with no signs of heat and the high voltage system has been disabled, the battery may be moved to a safe location.
3. Ensure the damaged battery or vehicle with damaged battery has a 50 foot buffer area around it from buildings or materials.

9. Important additional information

WARNING

Possible battery reignition! After a fire incident, store outside at a safe distance (50 FT/ 15 M) from other vehicles and structures!

WARNING

Always use appropriate rescue tools and always wear appropriate PPE. Failure to follow these instructions can result in serious injury or death.

WARNING

Regardless of the disabling procedure you use, ALWAYS ASSUME THAT ALL HIGH VOLTAGE COMPONENTS ARE ENERGIZED! Cutting, crushing, or touching high voltage components can result in serious injury or death.

WARNING

After deactivation, the high voltage circuit requires 2 minutes to de-energize.

WARNING

Handling a submerged vehicle without appropriate PPE for water rescue can result in serious injury or death.

WARNING

When fire is involved, consider the entire vehicle energized. Always wear full PPE, including a SCBA.

 WARNING

When cutting the first responder loop, double cut the loop to remove an entire section. This eliminates the risk of the cut wires accidentally reconnecting.

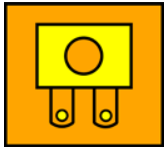
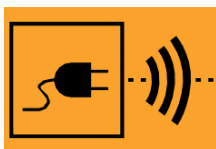
 WARNING

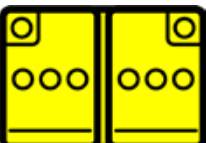
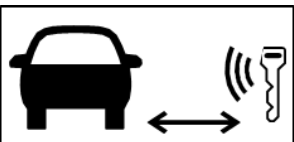
When using the high voltage shutdown methods recommended by this document, high voltage power is isolated to the battery. The high voltage battery is always energized.

Contact Us

First Responders and Second Responders with emergencies, call Kalmar Product Support at 1-785-229-6327.

The Kalmar Ottawa OT2 EV Owner's Manual and first responder information can be found at <https://www.kalmarottawa.com/en-US/parts-service/operator-manuals/>. First responders and training officers who have questions, contact kalmarusa@kalmarglobal.com.

10. Explanation of pictograms used	
	Battery Electric Vehicle
	High voltage power cable
	High voltage disconnect
	Fuse box disabling high voltage system
	Charge port
	Vehicle induction charging
	Cable cut
	Ultra capacitor, high voltage
	Battery, high voltage Lithium-Ion

	High voltage battery pack
	High voltage component
	Electricity / shock warning
	Shutdown hydrogen by closing the valve
	Ultra capacitor, low voltage
	SRS control unit
	Battery, low voltage
	Remove smart key /starter key
	Dangerous voltage

	Height control truck in driver cabin
	Steering wheel, tilt control
	Seat height adjustment by air system
	Seat adjustment, longitudinal
	Check battery temperature
	<i>In some working environments, the Infrared (IR) device is referred to as a Thermal Imaging Camera (TIC)</i>
	Lifting point; central support
	Flammable
	Acute toxicity

	Corrosive substances present
	Explosive
	Hazardous to human health
	Oxidizer
	Environmental hazard
	Emergency exit right hand
	Emergency exit left hand
	Emergency door opener
	Break window to obtain access

	Seat belt pretensioner
	Zone requiring special attention
	Left hand drive
	Fuel tank content Diesel
	Tank content oil
	Air tank
	Air-conditioning component
	Gas line (generic)
	Air-conditioning line

	General warning sign
	Warning; low temperature
	Use water to extinguish
	Use wet foam to extinguish the fire
	Use dry foam to extinguish the fire
	Use ABC powder to extinguish the fire
	Do not extinguish with water
	Special battery access
	Automatic fire suppression system